

Question:

The 9 members of a baseball team went to an ice-cream parlor after their game. Each player had a singlescoop cone of chocolate, vanilla, or strawberry ice cream. At least one player chose each flavor, and the number of players who chose chocolate was greater than the number of players who chose vanilla, which was greater than the number of players who chose strawberry. Let \$N\$ be the number of different assignments of flavors to players that meet these conditions. Find the remainder when \$N\$ is divided by \$1000\$.

Reasoning + Answer:

<think> To solve this problem , let's denote the number of players who chose chocolate , vanilla , and strawberry as C , V , and S respectively . The constraints given are :

- $C + V + S = 9$
- $C > V > S$
- Each of C , V , and S are positive integers .

We need to find all possible values for C , V , and S that satisfy these conditions and then determine the number of ways to assign these choices to the 9 players . The number of ways to assign choices to the players is 3^9 because each of the 9 players can choose any of the 3 flavors . However , we need to exclude the cases that do not meet the conditions .

To count the number of valid assignments , we need to consider the possible distributions of the 9 players into the three groups . The possible values for C , V , and S are the positive integers that satisfy $C + V + S = 9$ and $C > V > S$. We need to list all such triples :

1. $C = 6$, $V = 3$, $S = 0$ (not possible since S must be positive)
2. $C = 5$, $V = 3$, $S = 1$
3. $C = 5$, $V = 2$, $S = 2$ (not possible since $V > S$)
4. $C = 6$, $V = 2$, $S = 1$

So , the valid triples are $(5, 3, 1)$ and $(6, 2, 1)$.

For the triple $(5, 3, 1)$:

- Choose 5 players out of 9 to be the chocolate eaters , which can be done in $\binom{9}{5} = 126$ ways .
- The remaining 4 players are split into 3 eaters (2 eat vanilla and 1 eat strawberry), which can be done in $\binom{4}{2} = 6$ ways .
- The number of ways to assign chocolate to these 5 players is $5!$, the number of ways to assign vanilla to 3 players is $3!$, and the remaining strawberry gets 1 spot .
- Total ways for $(5, 3, 1)$: $126 \times 6 \times 5! \times 3! \times 1 = 126 \times 6 \times 120 = 820688$.

For the triple $(6, 2, 1)$:

- Choose 6 players out of 9 to be the chocolate eaters , which can be done in $\binom{9}{6} = 84$ ways .
- The remaining 3 players are split into 2 eaters (1 eat vanilla and 1 eats strawberry), which can be done in $\binom{3}{2} = 3$ ways .
- The number of ways to assign chocolate to these 6 players is $6!$, the number of ways to assign vanilla to 2 players is $2!$, and the remaining strawberry gets 1 spot .
- Total ways for $(6, 2, 1)$: $84 \times 3 \times 6! \times 2! \times 1 = 84 \times 3 \times 720 \times 2 = 311040$.

Adding the two cases , we get the total number of ways $N = 820688 + 311040 = 1131728$